

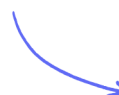
Structured Questions

Refraction, Reflection & Polarisation

Equation for the Intensity of Radiation / Refraction & Refractive Index / Critical Angle / Total Internal Reflection / Measuring Refractive Index / Converging & Diverging Lenses / Using Ray Diagrams / Power of a Lens / Thin Lenses in Combination / Real & Virtual Images / The Lens Equation / Magnification / Plane Polarisation

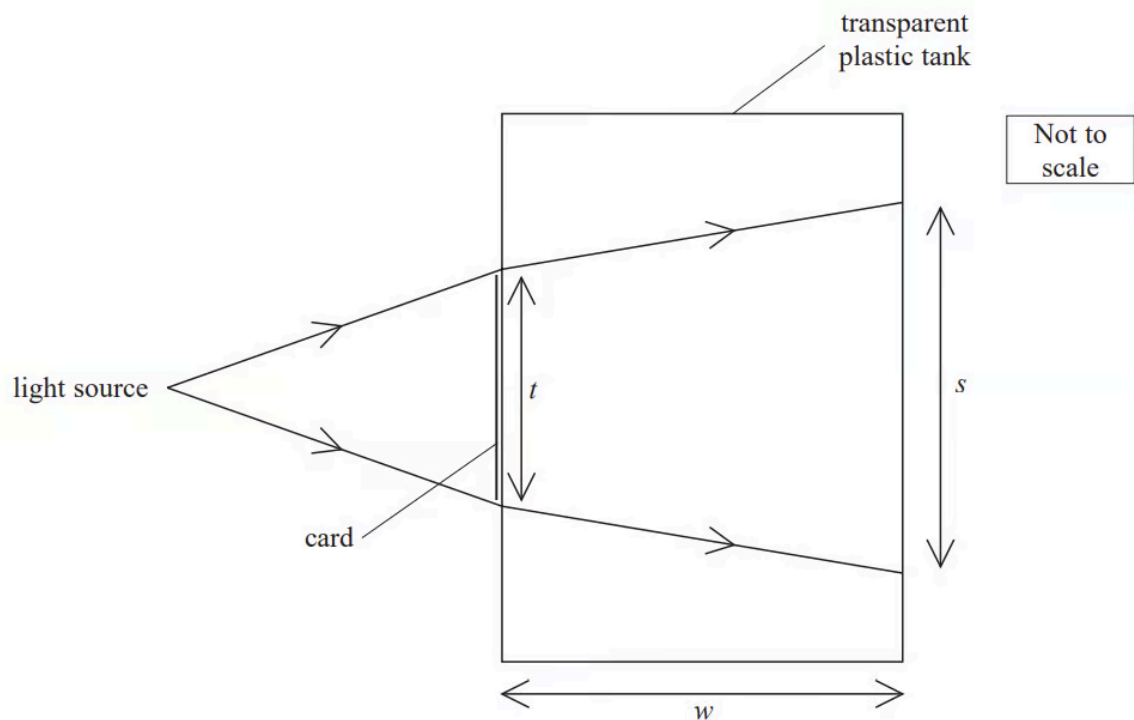
Medium (5 questions)	/39
Hard (1 question)	/5
Total Marks	/44

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Medium Questions

- 1 (a) The diagram shows a transparent tank, with thin plastic sides, that can be used to determine the refractive index of a transparent liquid.



A rectangle of opaque card is stuck on the side of the tank containing the liquid. A light source is placed in front of the tank and the width s of the shadow of the card, which is formed on the back of the tank, is measured. The width t of the card and the width w of the tank are also measured.

The angle of incidence of the light as it enters the tank is 7.2°

Show that the refractive index of the liquid is about 1.4

$$w = 35.0 \text{ cm}$$

$$t = 4.0 \text{ cm}$$

$$s = 10.2 \text{ cm}$$

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(3 marks)

(b) Determine the speed of light in the liquid.

Speed of light =

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(2 marks)

2 (a) The photograph shows a man wearing a virtual reality (VR) headset.

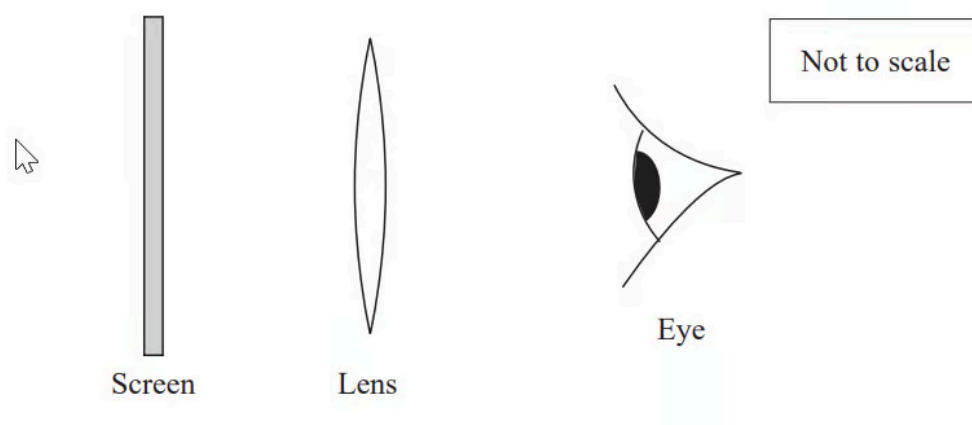


The VR headset gives the illusion of three-dimensional vision.

Inside the VR headset a pair of lenses is used to enable the user to focus on a magnified virtual image of a screen. The lenses can be changed to suit the vision of the user.



In the VR headset the lens is between the eye and the screen, as shown below.



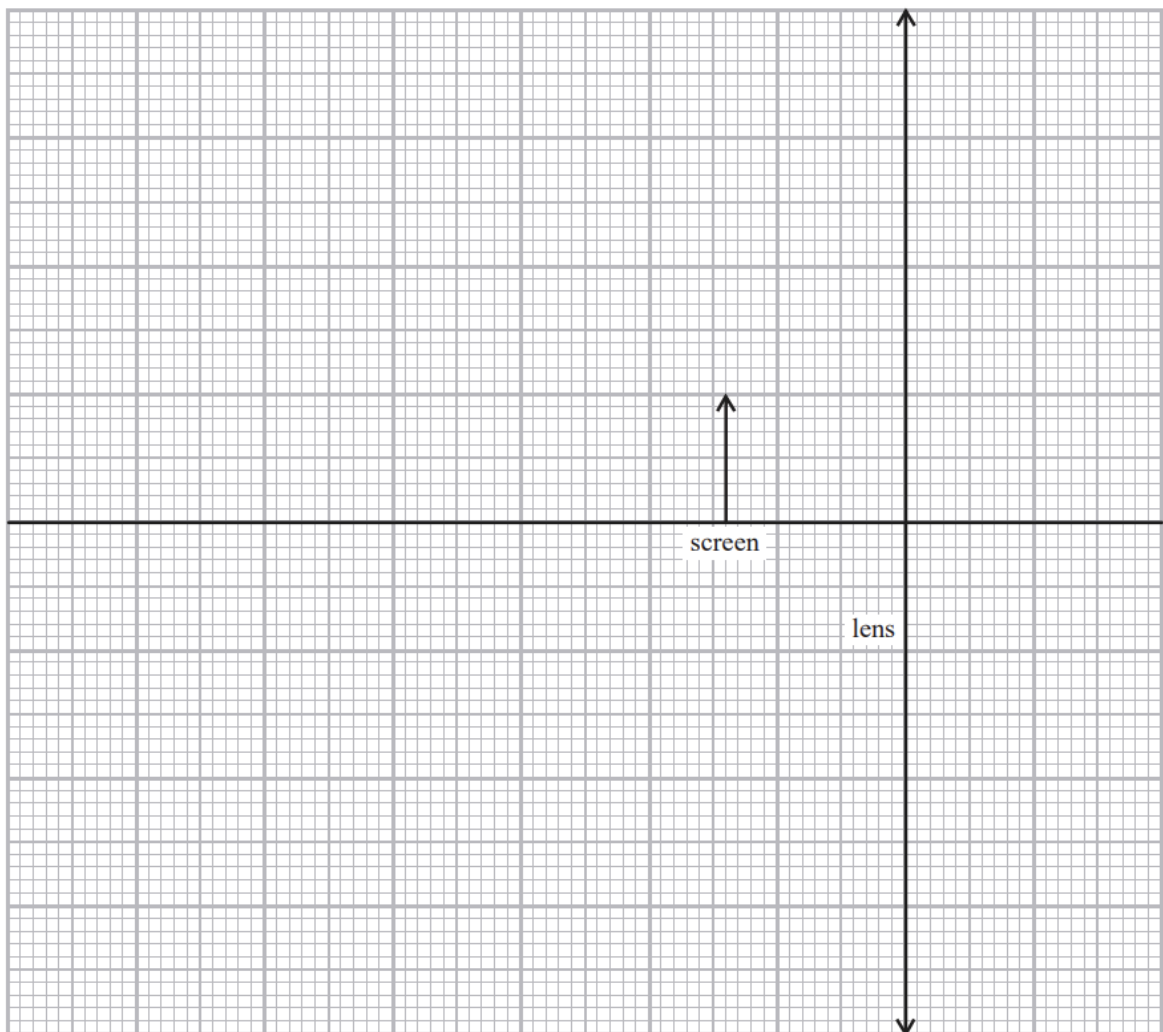
For a particular user of the headset, the image of the screen must be at least 16 cm from the eye and have a magnification of at least 3.0.

Determine whether this would be possible with a lens of focal length 3.8 cm.

Your answer should include a full-scale ray diagram drawn on the grid provided.

distance from screen to lens = 2.8 cm

distance from lens to eye = 2.2 cm



(4 marks)

- (b) Plastic Fresnel lenses are used in the VR headset because they are thinner and lighter than traditional glass lenses.

Instead of the continuous curved surface of a converging lens the Fresnel lens has circular ridges, each with an edge at a different angle to the adjacent ridge, as shown in

the simplified cross-section in Figure 1. Figure 2 shows a ray of light entering a section of the lens.

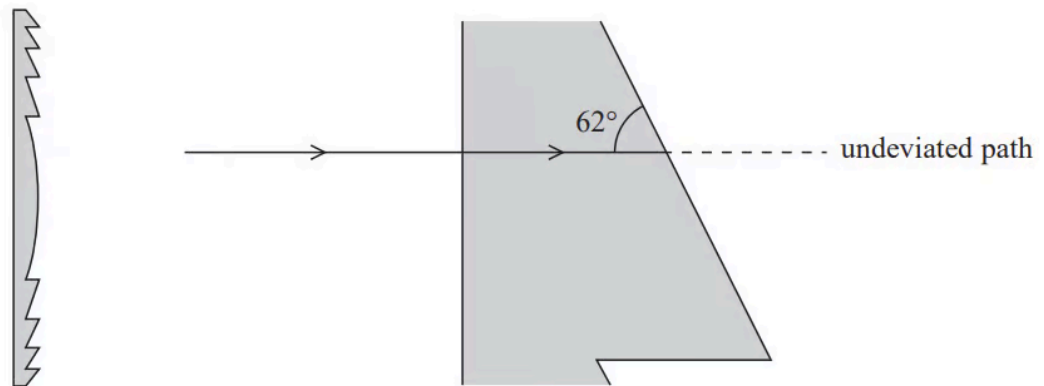


Figure 1	Figure 2
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i) Calculate the angle through which the ray has been deviated as it emerges from the plastic.

refractive index of plastic = 1.47

Angle =(4)

ii) Explain how the lens focuses a beam of light travelling parallel to the principal axis.

(3)

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(7 marks)

3 (a) A converging lens can be used to produce a real image on a screen.

A converging lens of focal length 15.0 cm is used to project an image of an illuminated object onto a screen. The object is a circle of diameter 4.0 mm and the image must be as large as possible on a screen of size 0.75 m by 1.25 m.

Calculate the distance between the lens and the screen for this image to be displayed.

Distance between lens and screen =

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(3 marks)

(b) A magazine article includes the statement:

If the distance from the lens to the screen is doubled, the brightness of the image is halved.

Assess the validity of this statement.

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(2 marks)

- 4 (a)** The photograph shows a sample of the mineral selenite. Selenite is made up of many long, narrow crystals.



Selenite has a refractive index of 1.52

Calculate the speed of light in selenite.

Speed of light in selenite =

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(2 marks)

- (b)** i) State what is meant by critical angle

(1)

ii) Calculate the critical angle for light in selenite.

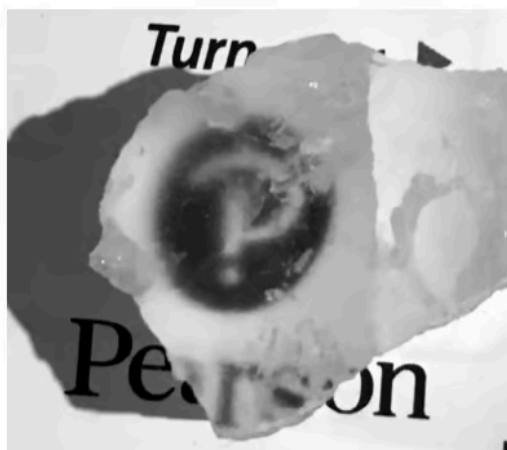
Critical angle for light in selenite =**(2)**

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(3 marks)

- (c) Selenite can act as a collection of optical fibres, so that an image of writing beneath the mineral sample appears as if it is at the upper surface as shown.



Explain how light travels through a selenite crystal.

(2 marks)

5 (a) The lens in the eye of an octopus focuses light onto the retina at the back of the eye.

The octopus focuses on objects at different distances from the eye by changing the shape of the eye to move the lens closer or further from the retina.

i) The power of an octopus lens is 118 D.

Show that the focal length of the lens is about 8.5 mm.

(2)

ii) Calculate the shortest distance from the eye at which an object may be focused clearly on the retina.

maximum distance from lens to retina = 2.0 cm

Shortest distance from the eye =**(2)**

iii) The lens in the eye of an octopus is in contact with seawater. The refractive index of freshwater is less than the refractive index of seawater.

Deduce what would happen to the shortest distance from the eye at which an object may be focused clearly if the octopus was in freshwater.

(3)

iv) Calculate the speed of light in seawater. refractive index of seawater = 1.37

Speed of light in seawater =**(2)**

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(9 marks)

(b) An octopus can detect the orientation of polarised light.

State what is meant by polarised light.

(2 marks)

Hard Questions

- 1 A student carried out an experiment to determine the focal length of a converging lens.

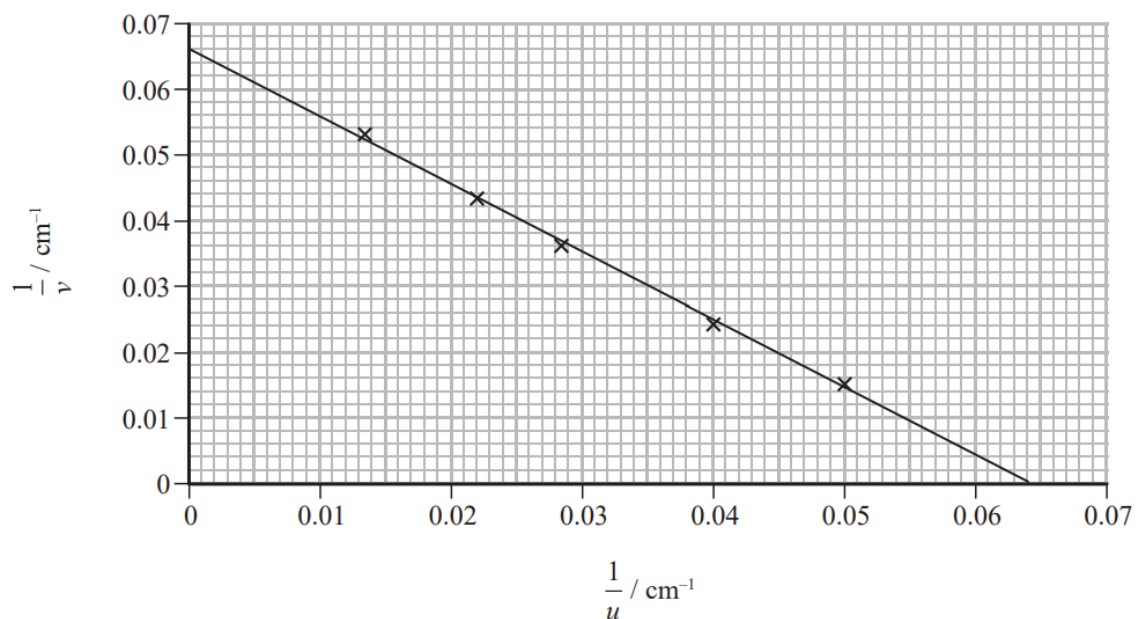
He placed the lens a distance u from an illuminated object. He placed a screen on the other side of the lens and moved the screen until a sharp image of the object was produced. He measured the corresponding image distance v .

The student repeated the procedure for four more values of u .

In his lab report he wrote:

"I made an initial determination of the focal length of the lens and concluded that it was about 15 cm. When I plotted a graph it confirmed my initial determination of the lens focal length."

The student's graph is shown.



Comment on whether the student's data is consistent with his initial determination of the focal length of the lens.

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(5 marks)